

HOMEWORK 4

>>NAME HERE<<

>>ID HERE<<

Instructions:

Use this LaTeX file as a template to develop your homework and submit it as a single PDF file on Gradescope. Remember to assign the pages to problems when submitting.

1 Verify PAC Bound

Let h be the VC dimension of function family F . For any $\delta > 0$, with probability at least $1 - \delta$ we have

$$R(\hat{f}_S) - \hat{R}_S(\hat{f}_S) \leq 2\sqrt{\frac{2}{n} \left(h \log n + h \log \frac{2e}{h} + \log \frac{2}{\delta} \right)}$$

where the log is natural, $R(f) = \mathbb{E}1_{[f(x) \neq y]}$ is the risk of f , $S = (x_1, y_1), \dots, (x_n, y_n)$ is a training set of size n , $\hat{R}_S(f) = \frac{1}{n} \sum_{i=1}^n 1_{[f(x_i) \neq y_i]}$ is the empirical risk of f on S , and $\hat{f}_S \in \arg \min_{f \in F} \hat{R}_S(f)$ is an empirical risk minimizer (ERM). We now verify the PAC bound on a simple classification task. Let $p(x) = \text{uniform}([-1, 1])$. Given x , the label is deterministic: $y = \text{sign}(x)$. Recall $\text{sign}(x) = 1$ if $x \geq 0$, and 0 otherwise. This means the true decision boundary is at $x = 0$. Let $f_\theta(x) := \text{sign}(x - \theta)$ which has threshold at θ . Let $F = \{f_\theta : \theta \in [-1, 1]\}$.

1.1 ERM [5 points]

Given S , suppose we find the smallest positive item: $a = \min_{(x_i, y_i) \in S: y_i = 1} x_i$ if one exists, otherwise let $a = 1$. Is $\hat{f}_S := f_a$ an ERM? Justify your answer.

[Your solution here.](#)

1.2 True risk [5 points]

Derive $R(f_a)$.

[Your solution here.](#)

1.3 Empirical risk [5 points]

Derive $\hat{R}_S(f_a)$.

[Your solution here.](#)

1.4 VC dimension [10 points]

What is the VC dimension of F ? Note F contains threshold classifiers of the type “left negative, right positive.”

[Your solution here.](#)

1.5 Generalization error [5 points]

Fix $\delta = 0.05$ (95% confidence) and $n = 200$. What is the bound on the true risk of $\hat{f}_S$ given by the PAC bound?

[Your solution here.](#)

1.6 Empirical verification [10 points]

Generate 10,000 random training sets from $p(x)$ and the associated labels. Each training set S has $n = 200$ points. On each S you will compute $R(f_a) - \hat{R}_S(f_a)$. Now you have 10,000 numbers: (1) produce a histogram of them, (2) find their 95% quantile of them, and (3) compare the 95% quantile to the bound in the previous question. Discuss your observations.

[Your solution here.](#)

2 VC dimension

2.1 Showing VC dimension [10 points]

Let the input $x \in \mathcal{X} = \mathbb{R}$. Consider $\mathcal{F} = \{f(x) = \text{sign}(ax^2 + bx + c) : a, b, c \in \mathbb{R}\}$, where $\text{sign}(z) = 1$ if $z \geq 0$, and 0 otherwise. Compute the VC-dimension $VC(\mathcal{F})$ of $\mathcal{F}$.

[Your solution here.](#)

2.2 Shatter Coefficients [10 points]

The main intuition behind VC theory is that, although a collection of classifiers may be infinite, using a finite set of training data to select a good rule effectively reduces the number of different classifiers we need to consider. We can measure the effective size of class $\mathcal{F}$ using *shatter coefficient*. Suppose we have a training set $D_n = \{(x_i, y_i)\}_{i=1}^n$ for a binary classification problem with labels $y_i = \{-1, +1\}$. Each classifier in $\mathcal{F}$ produces a binary label sequence

$$(f(x_1), \dots, f(x_n)) \in \{-1, +1\}^n$$

There are at most 2^n distinct sequences, but often not all sequences can be generated by functions in $\mathcal{F}$. Shatter coefficient $\mathcal{S}(\mathcal{F}, n)$ is defined as the maximum number of labeling sequences the class $\mathcal{F}$ induces over n training points in the feature space $\mathcal{X}$. More formally,

$$\mathcal{S}(\mathcal{F}, n) = \max_{x_1, \dots, x_n \in \mathcal{X}} \left| \left\{ (f(x_1), \dots, f(x_n)) \in \{-1, +1\}^n, f \in \mathcal{F} \right\} \right|$$

where $|\cdot|$ denotes the number of elements in the set. The *Vapnik-Chervonenkis (VC) dimension* $V(\mathcal{F})$ of a class $\mathcal{F}$ is defined as the largest integer k such that $\mathcal{S}(\mathcal{F}, k) = 2^k$.

Suppose there are n points $(x_1, \dots, x_n) \in \mathbb{R}$. Let $\mathcal{F}$ be the collection of 1-d linear classifiers: for each $t \in [0, 1]$, $f_t \in \mathcal{F}$ labels $x \leq t$ as -1 and $x > t$ as $+1$, or vice-versa. What is the shatter coefficient $\mathcal{S}(\mathcal{F}, n)$? What is VC-dimension $V(\mathcal{F})$? How can you get it from shatter coefficient?

[Your solution here.](#)

2.3 Monotonicity of VC-dimension [10 points]

Use the shatter coefficients to show the following monotonicity property of VC-dimension: if $\mathcal{F}$ and $\mathcal{F}'$ are hypothesis classes with $\mathcal{F} \subset \mathcal{F}'$, then the VC-dimension of $\mathcal{F}'$ is greater than or equal to that of $\mathcal{F}$.

[Your solution here.](#)

3 Kernel SVM

Consider the following kernel function defined over $z, z' \in Z$: $k(z, z') = \begin{cases} 1 & \text{if } z = z', \\ 0 & \text{otherwise.} \end{cases}$

3.1 PSD [10 points]

Prove that for any integer $m > 0$, any $z_1, \dots, z_m \in Z$, the $m \times m$ kernel matrix $K = [K_{ij}]$ is positive semi-definite, where $K_{ij} = k(z_i, z_j)$ for $i, j = 1 \dots m$. (Let us assume that for $i \neq j$, we have $z_i \neq z_j$.) Hint: An $m \times m$ matrix K is positive semi-definite if $\forall v \in \mathbb{R}^m, v^\top K v \geq 0$.

[Your solution here.](#)

3.2 Separability [10 points]

Given a training set $(z_1, y_1), \dots, (z_n, y_n)$ with binary labels, the dual SVM problem with the above kernel k will have parameters $\alpha_1, \dots, \alpha_n, b \in \mathbb{R}$. (Assume that for $i \neq j$, we have $z_i \neq z_j$.) The predictor for input z is

$$f(z) = \sum_{i=1}^n \alpha_i y_i k(z_i, z) + b.$$

Recall the prediction is $\text{sign}(f(z))$. Prove that there exists $\alpha_1, \dots, \alpha_n, b$ s.t. f correctly separates the training set.
[Your solution here.](#)

3.3 Prediction [10 points]

How does that f predict input z that is not in the training set?
[Your solution here.](#)